

Tawhidul Islam

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SUMMARY

Full-Stack Developer with hands-on experience building web and mobile applications, event-driven backends, and cloud architectures from the ground up. Passionate about AI integration, backend performance, and creating intuitive user experiences. Proven ability to design and ship end-to-end projects using modern frameworks and cloud technologies.

WORK EXPERIENCE

Full Stack Engineer Intern | *Zetheta Algorithms Private Limited (Remote)*

Aug 2026 – Sept 2026

- Engineered a production-grade, 8-step multi-step loan application form featuring real-time schema validation, conditional rendering, secure document uploads with client-side compression, and e-signature capture.
- Designed and implemented a custom API integration framework connecting enterprise SAP S/4HANA systems to a financial analytics platform using secure OAuth 2.0 protocols and robust ETL transformation logic.
- Architected a multi-region active-active disaster recovery framework across AWS regions for a payment gateway, incorporating Aurora PostgreSQL global replication and Kafka streaming to meet strict 99.99% uptime requirements.
- Developed an event-driven financial notification backend utilizing Apache Kafka and Redis to route 25+ real-time event types across 5 channels while enforcing TRAI DND compliance, automated retries, and rate limiting.

PROJECTS

Journey | *AI-Assisted Mobile Fitness Tracking Application*

Stack: *Flutter, Dart, Flask (Python), Supabase (PostgreSQL), OpenAI GPT API*

- Designed and built a cross-platform mobile app specializing in workout safety, form correction, and user commitment tracking.
- Integrated the GPT API to deliver real-time fitness guidance and automated form-correction feedback.
- Developed a lightweight Flask backend to manage data streams and bridge mobile actions with AI services.
- Architected a cloud database with Supabase (PostgreSQL) to store user profiles, tracking metrics, and historical progression data.

IntelliDoc | *AI-Powered Document Management System*

Stack: *React, Vite, FastAPI, Python, SQLAlchemy, Alembic, PostgreSQL, pgvector, AWS S3, BAAI/bge-m3,*

PyTorch

- Engineered a full-stack document management system enabling semantic, contextual file search using BAAI/bge-m3 embeddings and PyTorch.
- Implemented vector database storage with PostgreSQL + pgvector for efficient high-dimensional semantic queries.
- Designed a robust FastAPI backend managing database migrations (Alembic) and ORM via SQLAlchemy.
- Built a fast, responsive frontend with React and Vite for optimized build times and smooth user interactions.

Full-Stack CRUD Web Application | FullStack Project

Stack: JavaScript (ES6), HTML5, CSS3, REST APIs, Node.js, PostgreSQL

- Built a complete full-stack web application with a clean client/server separation for scalable architecture.
- Developed a dynamic, responsive UI using vanilla JavaScript, HTML5, and CSS3 for real-time data management.
- Created RESTful server-side routes and API endpoints connecting frontend interactions to backend logic.

TECHNICALSKILLS

Languages: C++, Java, Python, Dart, JavaScript (ES6), TypeScript, SQL/MySQL

Web & Frameworks: React, Vite, Node.js, Express, Flask, FastAPI, HTML5, CSS3, REST APIs

Databases: PostgreSQL, Supabase, pgvector, SQLAlchemy, Alembic, Redis, Apache Kafka

Tools & Platforms: AWS (S3, Architecture), Firebase, Google Cloud Platform (GCP), Docker, Git, GitHub/Gitlab, PyTorch, OpenAI API

EDUCATION

B.A. in Computer Science | Hunter College, CUNY Sept 2023 – May 2026 *Relevant Coursework: Software Engineering · Data Structures · Computer Architecture · Operating Systems · Computer Theory · Advanced Programming Techniques*

A.S. in Computer Science | LaGuardia Community College, CUNY Sept 2021 – Aug 2023
Coursework: Intro to Computer Science · Discrete Structures · Calculus · Object-Oriented Programming